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Research paper

Interconnected digital twins of subway stations for decision support systems

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ABSTRACT

This article examines a digital twin (DT) for complex infrastructure assets using the subway system as a case study. The proposed solution includes twins for each station, which are interconnected to interact during various operational scenarios. Existing digital twins for subway systems are examined, and their shortcomings—primarily the latency in operator information retrieval due to slow model rebuilding—are identified. The study addresses decision support during medium- and large-scale incidents requiring the synchronization of multiple departments. The authors propose a digital twin architecture based on data collected from cameras, microphones, and sensors, processed within a fog computing environment. A separate digital twin is developed for each station, capable of utilizing templates from previously built assets. An accompanying mobile application with an intelligent assistant displays detected events and answers queries via voice interaction. Implementation results indicate that personnel response time to emerging events improved by 17.4% compared to the REPEAT platform. The proposed architecture is applicable to subway systems and other public transportation networks requiring coordinated personnel actions during hazardous situations.

1. Introduction

A digital twin (DT) is a digital model of a real-world object. Modern DTs are not only maps, they also support usage of situational modeling, simulation, and other tools that allow for the display and evaluation of the states of simulated objects in the present, past, and future. Fire alarm systems are especially in demand during hazardous situations, where decisions must be made appropriately within the available time and resources.

Particularly in hazardous situations (fire or explosion threat), a fire alarm system can help coordinate personnel and activate a remote fire suppression system to eliminate the emergency without direct human intervention, especially in cases where the danger to humans is extremely high (e.g., risk of death for a firefighter or chemical engineer).

A fire causes significant economic and human losses, but flooding from a fire can also cause significant damage to a building, and in some cases, lead to its partial destruction or the destruction of specialized equipment, machinery, etc. In each case, activating an automatic fire suppression system and flooding the premises with carbon dioxide will lead to high expense, and in some cases, the use of automatic fire suppression may require the construction of a new building or major renovations.

Precise and well-considered personnel actions and the use of intelligent systems will help minimize damage to an enterprise or organization. Recent advancements have demonstrated the efficacy of machine learning and intelligent systems in managing specific subway components, such as escalator passenger safety [1–3] and intelligent video surveillance scripting [4]. Despite these component-level successes, comprehensive digital twins for entire stations face significant limitations. Currently, digital twins are used in many subject areas, in particular, in the petrochemical industry, subways, railways and tunnels, where even a minor error can cost lives and lead to significant financial losses.

However, existing digital twins have a number of significant limitations, in particular:

1. there is no ability to process video images, sounds and data from sensors installed at facilities and there is no ability to promptly detect events at facilities, including dangerous ones, which does not allow for their prompt elimination;

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2. there is no mobile platform (smartphone application) with support for devices of the Internet of Things stack, which leads to a slow response to events;
3. there is no support for cloud distributed computing and fog computing environments, which leads to delays in data processing;
4. a web interface for work through a browser and from mobile devices (responsive website design) has not been developed, which causes difficulties in accessing the systems from any device and OS;
5. integration with intelligent video surveillance systems (IVSS) is not supported, preventing the prompt acquisition of images of ongoing events;
6. an intelligent assistant with AI-based voice assistant capabilities is lacking;
7. the ability to view a map of facilities and connect to video cameras in real time, display video footage, and detect events at subway facilities is lacking.

In the subway sector, these limitations hinder prompt decision-making and reduce the response time of personnel and support services at subway facilities. The required subway twin must possess new capabilities and be built on new principles. The new digital display for the subway must display the actual simulated facility. It must also allow events to be displayed on a map (2D) and in three-dimensional space (3D). The displayed model of the facility must have multiple nesting levels, be navigable using a mouse, keys, or other pointers, and display various fragments of the observed facility, including small fragments, in detail. It must also be possible to zoom in and out of the image in vector format.

The digital twin should be capable of playing video clips and audio files, giving operators a clearer understanding of situations occurring at the facility. Typical examples of digital twin use include the following: The system detects a “scream”, the operator selects the “play” function and listens to the clip. The system identifies the event at the facility as a “fire”, and the operator watches the video clip and reclassifies it as “smoke”.

To conveniently work with the digital twin, operators need an intelligent assistant with functions for displaying objects and situations arising within them, as well as speech recognition, which can answer operator questions about events occurring in the subway. Furthermore, the assistant’s speech recognition capabilities will allow navigation through the digital twin objects displayed on the screen using voice, which is convenient for all operators in dark and poorly lit subway areas.

The main challenges in creating the considered digital twins are determined by the complexity of the structure and behavior of subway objects, the highly dynamic changes occurring within the subway, and the large volume of data collected by cameras, various sensors, and microphones installed in stations and other subway facilities. Furthermore, the development of intelligent assistants for operators of complex infrastructure facilities, where hazardous situations can occur that require rapid personnel coordination, is poorly worked out. Another significant problem is the high cost of creating twins and upgrading them.

The authors propose creating a digital twin with a two-level structure: the upper level will house the digital twin of the subway, while the lower level will house the digital twins of individual stations. The proposed digital twin is scalable, with intermediate levels being created where stations are grouped together. For example, groupings could be based on districts, lines, and station types: indoor/outdoor, single-platform, transfer, multiple-exit, with or without escalators. It is possible to use existing digital twins of individual stations if they were previously built.

For new stations, new digital twins are proposed to be built using a specially developed template, which is a standard digital twin. When a new digital twin is built using this template, the standard digital twin is modified taking into account the station’s specific features, as well as previously developed digital twins for other stations. In particular, when constructing a subway station’s digital twin, the template varies depending on the architectural plan of the specific station: non-standard exits, passages, multi-level escalators, etc.

This study addresses the identified gaps by making the following primary contributions:

- The proposition of an interconnected digital twin architecture linking neighboring stations for preemptive response.
- Integration of a fog-computing environment to process multi-modal data, significantly reducing latency.
- Development of a mobile intelligent assistant for real-time $2D/3D$ visualization and voice-navigated management.

The following section considers popular digital twins. In Section 3 intelligent assistants are analyzed. The research statement is formulated in Section 4. General description of the proposed solution is presented in Section 5.1, digital twin construction platform is discussed in Section 5.2, standard digital twin template is considered in Section 5.3 and digital twin implementation for the subway is described in Section 5.4. Section 6 contains the example of use. Findings and conclusion are in Section 7.

2. Popular DTs

While commercial platforms dominate industrial applications, recent academic literature has heavily focused on the theoretical frameworks of digital twins in transportation and infrastructure. Broader applications of DTs have been proposed for building construction frameworks [5], condition assessment of railway infrastructures [6], arch cover method monitoring [7], rail operation safety assessment [8], and maintaining resilient ports [9]. Specific to subway environments, Kaewunruen et al. [10] explored sustainability audits for subway stations, and Shevlyugin et al. [11] modeled traction power systems. More recently, Zhao et al. [12] and Faraji et al. [13] highlighted the critical role of DTs in subway flood risk management and intelligent tunnel monitoring. Despite these advancements, a gap remains in integrating real-time, fog-assisted multi-modal sensor fusion [14] into a unified, mobile-accessible framework for station operators. The vast majority of existing commercial tools for developing digital twins.

A popular tool for creating interactive diagrams and digital twins is the REPEAT platform. REPEAT’s advantages are the ability to develop highly complex models of technological objects (nuclear reactors, aircraft, electric vehicles); the ability to create and configure a digital twin control system; the ability to conduct multiple virtual tests; and the ability to generate control algorithm code for the target device. The platform is intended for use in design institutes, engineering bureaus, and educational institutions. The REPEAT platform supports cloud infrastructure with flexible access via a web browser from any device; data integration; model integration; and a built-in parallel computing library. The REPEAT platform is used by Rosatom and many other technology giants for building digital twins.

One widely used system for displaying monitored objects on maps is Milestone XProtect. The system is used for video surveillance and allows viewing event logs and pattern recognition. It also features a map designer for designing maps of monitored objects. Maps are managed in the Management Client 2020 R1 subprogram from the XProtect program’s main menu. Smart maps in XProtect Smart Client do not reflect the actual behavior of the monitored object and do not include simulation tools.

Other popular tools for developing digital twins in the world are: Ansys Twin Builder, COMSOL Multiphysics, CML-Bench, Unreal Engine, SOLIDWORKS Simulation, FlowVision, Rocky DEM, AMBER, SimManager, Altair: One Total Twin platform, Microsoft: Azure Digital Twins, AnyLogic, iTwin, PlantSight, and Merius Oy Smart Mill.

Ansys is a multimedia platform that provides sets of elements and connections for connecting to sensors. The initially large element library can be expanded, and models can be built and displayed in detailed layer-by-layer manner. Ansys supports the creation of prototypes

of large systems with optimization of various characteristics for operation on a real object. Another popular program for building process diagrams, COMSOL, allows to build process diagrams and analyze individual interrelated processes. It allows to define material properties and process physics, and COMSOL is suitable for almost any user.

CML-Bench supports mathematical modeling and digital design capabilities. It has its own core and database for calculations, and enables comparison of modeling results with models created using 2D modeling systems. Models are stored in files (all popular CAD system standards are supported): AutoCAD (.dwg), SolidWorks (.sldprt, .sldasm), Inventor (.ipt, .iam), CATIA (.CATPart, .CATProduct), STEP (.stp, .step), IGES (.igs, .iges), and STL (.stl).

Unreal Engine works well with planning functions and supports importing models from all popular CAD packages. SOLIDWORKS Simulation breaks the model into finite nested models. Typical models of the program include: biological cell growth and heat transfer. It is possible to quantitatively evaluate physical phenomena occurring in liquids and solids, as well as radio wave propagation, using the finite element method and other numerical methods.

FlowVision is used to study flow processes in fluid dynamics. Methods of the physical impact of fluid flow and heat transfer on the modeled object can be used. The model generated by the program can be verified taking into account chemical processes occurring in a real object (flasks, simulation rig) using the simulation model. Physical prototyping of a soil environment with soil (aggregate) destruction is possible on a simulation rig (physical object) and comparison with a digital model based on numerical methods.

Unlike FlowVision, Rocky DEM builds a model based on discrete elements and allows for the simulation of soil environment destruction. A model template for the early stage of solid deformation is provided, which is necessary when modeling rocks and ice (fluid mechanics). The topology used includes material particles and rock blocks. The motion of individual model elements and their links with the deformated object can be determined. Together, the deformated object and the model elements form a comprehensive soil environment model with a fixed starting point and conditions for the motion of the model elements.

AMBER is designed for modeling chemical processes at the molecular level. The SimManager program simulates biological processes and object interactions: models of living organs (kidneys, liver, spleen, etc.) with the environment as a set of discrete particles, with the ability to specify a smoothing length (return to the initial state before impacting the organ). Operation in liquid environments is possible, with the use of kernel smoothing and kernel approximation functions.

The Altair software platform allows the development and launch of digital twins in any environment: in a browser, on a smartphone, or on a computer. All data is stored in the cloud. The platform provides machine learning tools and neural networks used to build models of real-world objects. The Azure Digital Twins cloud platform also provides the ability to host applications in various data centers. The Azure environment includes models of complex environments, including subways, railways, and entire cities. The platform is positioned as a platform for building a business in a way that reduces costs and optimizes operating expenses. The platform enables companies to analyze models, improve customer service, and enhance their products.

AnyLogic is a simple, fast, and popular simulation environment suitable for all industries. Models built in AnyLogic are highly flexible, applicable to a variety of tasks, and can be reconfigured based on input data. iTwin allows for modeling and visualization of infrastructure assets with the ability to convert them to a BIM model. The program supports a hybrid 2D/3D design environment with timeline visualization and data completeness verification. iTwin allows for the creation of interactive monitoring models. The program's primary purpose is to implement engineering projects.

PlantSight allows for the construction of economic assets in 2D/3D space and the description of various interactions between objects to develop problem-solving strategies, regardless of the physical location of the assets. The program supports Industrial Internet of Things (IIoT) technologies with augmented reality elements. The first version of PlantSight's digital twin provided the ability to build enterprise models, and later, models of smaller physical assets were developed.

Merius Oy Smart Mill uses the ContextCapture engine, which creates 3D models. A distinctive feature of the platform is its ability to utilize drone images, laser scanning data, and visualize information in real time in a 3D environment.

The vast majority of existing tools for developing digital twins include graphic elements that are both part of the digital twin development environment and elements of the digital twin itself (people, buildings, workshops, entrances, parking lots, etc.). It should be noted that many digital twins operate only within the digital twin development environment and do not have the ability to be launched separately (*.flv, *.exe, *.class, *.app, etc.).

Currently, in the situation center of the State Unitary Enterprise "Petersburg Subway" Milestone XProtect video surveillance system is used as a monitoring and decision support system. Events that may occur in the subway and should be monitored by staff are structured into different categories. An example of such events is the discovery of abandoned items by passengers. Video clips showing these events are copied to a separate secure server. Video streams from all cameras are stored in the city's shared storage for 30 days. Servers optimized for working with large volumes of data are located at the City Monitoring Center and are accessible through the "Safe City" program, where Milestone XProtect, VideoNet, and other programs are used.

Digital twins of subway stations are part of the software platform and development environment of the digital twins. Table 1 presents the results of a comparison of existing systems. The capabilities of existing systems were compared with those that must be supported by a system designed to create digital twins for subway stations.

The following notations are used in Table 1:

- **p1** – support for a mobile platform and IoT devices;
- **p2** – support for distributed computing in a fog computing environment or cloud platform;
- **p3** – availability of a web interface for working via a browser on mobile devices (responsive website design);
- **p4** – integration of the digital twin model into the intelligent video surveillance system;
- **p5** – AI-based voice assistant;
- **p6** – viewing a map of objects and the ability to connect to video cameras in real time;
- **p7** – processing data from sensors, sounds, and video images with the detection of events at subway facilities and the ability to display videos.

All systems for building digital twins have their own specific features and are designed to model specific objects (physical, economic, material, chemical, geological, etc.) in a specific subject area (railways, airports, loading terminals, nuclear power plants, metal smelting, etc.) and to achieve specific results (reducing business costs, calculating physical quantities, predicting object behavior, using a simulation timeline - changing the model state over time).

Each system (platform) for digital twin design has undeniable advantages in one application area (domain) when solving one or a limited set of specialized tasks and can hardly be used to build digital twins of the subway.

Existing general-purpose software does not meet the following requirements: existence of an application for working with the digital twin for a mobile platform (smartphones, Internet of Things devices), support for cloud and fog computing, the existence of a website (adaptive interface for all devices), integration with intelligent video surveillance systems, existence of an AI-based voice assistant; possibility to view a map of subway facilities with the ability to connect to video cameras, processing data from sensors and various devices, that does not allow to create subway digital twin that meet modern requirements.

Table 1
Comparison of existing systems.

Systems	p1	p2	p3	p4	p5	p6	p7
REPEAT	–	+	+	–	+	–	+
Milestone XProtect	+	–	+	+	–	+	+
Ansys	–	–	–	–	–	–	+
COMSOL	–	–	+	–	–	–	–
CML-Bench	+	+	+	–	+	–	–
Unreal Engine	–	–	+	–	–	–	+
SOLIDWORKS Sim	–	–	–	–	–	+	+
FlowVision	–	–	–	–	–	+	+
Rocky DEM	–	–	–	+	–	+	+
AMBER	–	–	–	–	–	+	+
SimManager	–	–	–	–	–	+	–
Altair	+	+	+	–	–	–	–
Azure Digital Twins	–	+	+	–	–	–	–
AnyLogic	–	–	+	–	+	–	–
iTwin	–	+	+	+	–	–	–
PlantSight	+	+	+	+	–	–	+
Merius Oy Smart Mill	+	+	+	+	–	–	–
Ansys Twin Builder	–	+	–	–	–	–	+

3. Overview of intelligent assistants

Currently, the most widely used intelligent assistants include the following: Alisa, Google Assistant, ClickUp AI, Copy.ai, Amazon Alexa, Siri, Notion AI, Zapier AI, Jasper Generative AI Chatbot, SaneBox, MindPal, and other programs that are virtual assistants with artificial intelligence. Some intelligent assistants have highly accurate voice recognition models in national languages (English, Russian, etc.) and can be integrated into smart devices.

One such assistant is the Alisa assistant from Yandex. This intelligent assistant not only includes a smart speaker but also provides functionality that requires interaction with other devices such as purchasing additional services via a smartphone or PC, sending emails. Alice supports integration with Smart Home sensors, can turn lights on and off, open and close windows, operate an intercom, recognize faces, act as a search engine, and provide other services.

The voice assistant from Google (Google Assistant) can maintain a conversation, search for information online and return search results, provide weather information, create vehicle routes, perform simple mathematical operations, and convert various values, such as exchange rates and measurements. The Assistant app can translate phrases into another languages (from English to Spanish, from Chinese to Japanese, etc.), search for recipes for homemakers, provide sports scores, explain word meanings, and display exchange rates. The app can work in standby mode and runs on all popular operating systems: iOS/iPadOS/macOS, Android, Linux, and Windows.

ClickUp AI is not only a voice assistant but also a project management service based on the Kanban methodology, allowing to adjust workflows, visualize all project stages, and prioritize their completion. Compared to Google Assistant, which is more focused on home use, ClickUp's voice assistant is oriented on solving business problems. ClickUp's modules allow to manage tasks and assign roles, while its artificial intelligence (AI) module generates technical specifications, reviews checklists, and creates subtasks.

Amazon's Alexa voice assistant is designed primarily for home use and works best with Amazon Echo and Amazon Echo Dot speakers. Alexa supports virtual assistant functions, including music playback and audiobooks reading. Alexa can create to-do lists, set alarms, leave voice messages, and provide news updates, stock quotes, sports scores, weather, and traffic information. Like Yandex's Alisa, Alexa can control smart home devices (opening doors, windows, turning on lights, etc.). Alexa supports multilingual dialogue in Japanese, Spanish, German, and other languages—not just English.

Siri is the voice assistant built into Apple operating systems. Siri can make calls (on iPhone or iPad) and launch apps on all devices: iPhone, iPad, Mac, Apple Watch, and Apple TV. Siri can send messages in chats, manage alarms, calendars, and to-do lists, build routes, search the internet, get news and weather forecasts, play music, and search for movies based on user requests. In the latest version of macOS 15.2 Sequoia, Siri can generate images using artificial intelligence algorithms and remember conversations. Siri can run in standby mode on MacBooks using Apple Intelligence technology.

Notion AI is designed to organize information in a unified workspace, offering both business and home use. Notion AI stores information in the form of notes, abstract documents, and files in the cloud. Notion AI allows to organize data in tables and dashboards, create calendar events, and manage projects. The key feature of Notion AI is its modularity. It allows to add modules for working with tables, lists, databases, message boards, and galleries. Notion AI can be accessed through a browser or through a desktop app, both of which can be installed on popular operating systems: MacOS, Windows, Android, Linux.

Zapier AI, an AI-powered automated assistant, combines the functionality of Slack, Trello, and Gmail so the user never misses important events. Zapier's core elements are: Zap (automated workflow), Trigger (process launch), and Action (actions that occur after the trigger is triggered). Notifications in Zapier are sent to the Slack module, which is designed for user interaction. Data in Zapier is stored in Smartsheet tables, which must be created before using the assistant. Zapier offers integration with email and social media. Zapier also allows to automate business processes.

The Jasper Generative AI Chatbot is an alternative to ChatGPT with text generation capabilities. Jasper can work with blogs, social media, and email. Jasper is suitable for business and can generate images. Jasper uses the same Generative Pre-trained Transformer language (GPT) model as GPT-3.5. Jasper AI supports integration with Chrome and other programs. Jasper AI can write both books and informational guides.

SaneBox provides email as a service, manages accounts, and functions as an intelligent assistant. SaneBox works with all email services. Its main function is to calculate email statistics for several days and delete unnecessary emails over time, using artificial intelligence algorithms. SaneBox supports IMAP and Exchange Web Services.

The MindPal assistant is focused on education and business. It enables the creation of artificial intelligence agents to improve company productivity and automate business tasks. MindPal offers analytics, task management, statistics visualization, advertising auditing, business plan generation, and production improvement proposals.

Table 2 provides a comparative analysis of the capabilities of existing intelligent assistants. The systems were compared based on the features required for the intelligent assistant for subway employees.

The table uses the following notations:

- **f1** – voice chatbot tailored to the specifics of the subway;
- **f2** – information about recognized objects at the subway stations;
- **f3** – information about emerging situations, taking into account data from other stations;
- **f4** – speech recognition;
- **f5** – sending text messages for the operator;
- **f6** – deep integration with the IVSS, video cameras, and end-user devices;
- **f7** – the ability to navigate objects displayed in the digital twin.

All existing assistants are digital assistants designed to solve specific tasks (for home, business, planning, accounting, and education), and help to reduce costs, save time at home, reduce stress, and so on. Each assistant has its own activity profile, defining the area in which the digital assistant operates most effectively. Currently, there are no digital assistants that offer all the functions required of an intelligent assistant for personnel working in complex infrastructure facilities.

Table 2
Required functions for intelligent assistants in the subway.

Name	f1	f2	f3	f4	f5	f6	f7
Alisa	–	–	–	+	+	+	+
Google Assistant	–	+	–	+	+	+	+
ClickUp AI, Copy.ai	–	–	+	+	+	–	–
Amazon Alexa	–	+	–	+	+	+	+
Siri	–	+	–	+	+	+	+
Notion AI	–	–	+	+	+	–	–
Zapier AI	–	–	+	+	+	–	–
Jasper Gen. AI	–	+	+	–	+	–	–
SaneBox	–	–	–	+	+	–	–
MindPal	+	–	+	+	+	–	–

4. Statement of the problem

The conducted analysis has shown that there are currently no digital twins that ensure high speed of receiving data on events as they occur in subway systems and ensure coordination of personnel actions aimed at eliminating hazardous and emergency situations and their consequences. It is necessary to create a new digital twin with significantly broader capabilities: the ability to process video data, audio data and data from various sensors that is continuously received from subway facilities, the ability to take into account the relationships between stations and the impact of events occurring at one or more stations on other stations.

For end users, it is necessary to create an intelligent assistant that is installed on user devices and provides visualization of events on 2D/3D models with varying degrees of detail, has speech recognition capabilities, which allows for the provision of information on recognized objects, receiving information on emerging situations, asking questions verbally, sending text messages to the operator; that is able to navigate in the digital twin's graphical objects, and also has the ability to integrate with the IVSS.

To ensure the required capabilities, the digital twin must include the digital twins of individual stations, and the digital twins of adjacent stations must be interconnected. It must be possible to use existing digital twins and create new twins using a template. When creating digital twins based on a template, the template-based digital twins must be adapted to the specific characteristics of the specific station. Furthermore, knowledge gained from the construction and use of digital twins of other stations must be taken into account.

Creation of digital twins of the stations should allow fast identification of emerging events (fire, flooding, littering, etc.) based on data collected from subway facilities, predict their progression, assess the potential impact on adjacent upstream and downstream stations, evaluate potential consequences, and quickly inform station personnel of any subway incidents.

The digital twin should include two components: a basic component capable of solving problems requiring the use of AI, and a graphical component providing 2D/3D models of subway stations used to display events on user devices. The digital twin should be built using sensor data, video information, and audio data processed using neural networks and should be also capable of solving tasks such as event detection, forecasting, and other tasks that require intellectual capabilities. Neural networks are used since they allow accurately identify objects and events thanks to more complex algorithms, though they are more demanding on GPUs [15].

Digital twin computationally complex processes should be executed in a fog computing environment on a server (as they are resource-intensive) adapted for processing data using neural network methods. Digital twins can be designed using ready-made platforms with adaptive elements developed for a specific subject area (domain) and supporting the connection of signaling elements (sensors, video cameras, audio, etc.). However, creating custom digital twins and platforms for their design will significantly improve the performance of systems that use these custom digital twins, compared to systems that use digital twins built using existing large platforms.

Custom digital twins and platforms operate several times faster because they don't have a large number of universal functions and elements that are unnecessary when creating subway digital twins. Creating a custom digital twin also allows for more precise customization of the digital twin based on data sources. Furthermore, using almost all standard digital twins poses installation and maintenance challenges, and they also have limitations in supported elements, objects, movements, and model dimensions.

When developing platforms for building digital twins, many development environments can be used, providing ready-made libraries and components for drawing graphics, drawings, diagrams, and so on. All elements or objects of digital twin (people, cars, trains, etc.) can be drawn using these libraries. It is also possible to create diagrams in 2D space, program the reception of signals (data from sensors, audio, video, etc.), and create your own digital twin in a programming language. The creation process may slow down by 30–100 times, but the result digital twin will be faster than digital twins that are built using a standard platform.

Developing a digital twin for the subway should involve developing components for processing data received from various sources installed at subway facilities, which are used to build the digital twin, as well as developing a mobile application for subway employees, which implements the functions of an intelligent assistant and uses the developed digital twin as a model. The data processing component should process continuously incoming data streams using neural networks.

The component should be hosted on a server in a fog computing environment, and data on detected events should be sent to intelligent assistants installed on staff mobile phones. It should be possible to install the intelligent assistant on various Android smartphone versions, as most devices worldwide run this operating system. It is necessary to create an APK application for a stable version of Android to avoid the need for employees to change devices.

The Android application can reside permanently in the smartphone's memory and display events in the status bar. After the initial connection to the fog server via HTTP(s), a persistent connection is established (TCP/IP route), which is very fast and can transmit event data in fractions of a second (10–200 ms). If the fog server is located in a local Ethernet network without internet access, a high-speed connection can be maintained for years.

Fig. 1 shows the composition and interaction diagram of the components of a decision support system based on the construction and use of a digital twin of the subway and interconnected digital twins of individual stations. The system includes a server application and an intelligent assistant (Fig. 1, left). The user application connects to the fog server via Wi-Fi. The fog server has expansion boards connected to devices such as microphones, video cameras, and sensors. The information on the server is processed by neural networks. Detected events and associated video and audio fragments are available for viewing from a mobile app. The Fog Server in the mobile app changes based on GPS coordinates and displays the subway station closest to the operator (the phone running the Fog Server app). The intelligent assistant can answer questions about train arrival times and many other issues related to the subway and conditions at nearby stations.

Creating a digital twin of a subway system for a decision support system presents a significant number of challenges. Difficulties arise when connecting devices and processing data on the fog server, which the developer must organize independently. Specifically, in some cases, it is necessary to develop drivers and application-device interaction functions (custom functions, in addition to OS kernel functions). Furthermore, difficulties arise when creating an intelligent assistant. Not all models used in intelligent assistants can accurately recognize speech. Speech synthesized using speech synthesis algorithms often contains sharp changes in pitch, inconsistencies, and unsteadiness.

Another challenge is displaying information transmitted from the server, containing image and event recognition results on a map or in

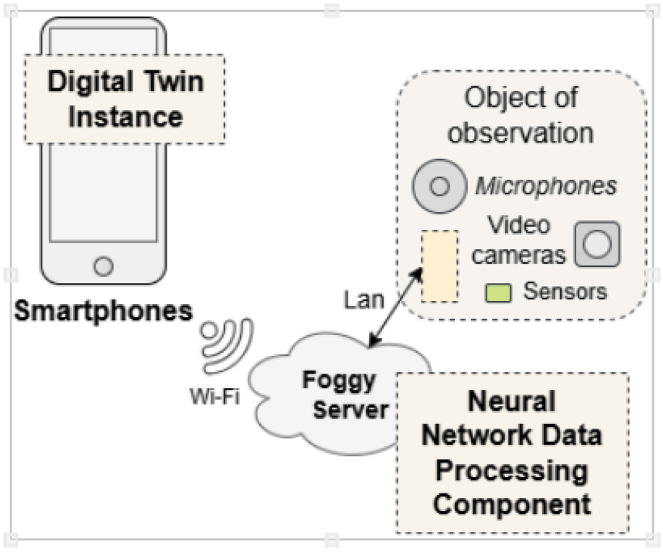


Fig. 1. Composition and interaction diagram of the components of a decision support system based on the construction and use of a digital twin of a subway system and interconnected digital twins of individual stations.

3D space in a mobile app. Thus, creating a custom digital twin requires developing drivers for non-standard devices and functions for receiving and transmitting data to a fog server, ensuring data processing by neural networks, and displaying information received from the server in a mobile app on a smartphone. It is also necessary to identify tools to ensure accurate speech recognition and synthesis.

The new subway digital twin and intelligent assistant for mobile phones have the following distinctive features:

1. **Digital twin allows to Increase speed of work of subway employees.** The use of PCs with monitors by subway operators significantly limits employee mobility, as employees are located in a secure center and spend significant time moving to the scene of an incident.
2. **The intelligent assistant supports speech recognition**, allowing employees to use their voice to navigate a graphical model displayed on the smartphone, helping them better understand the situation at the station. The speech recognition feature also allows employees to ask the intelligent assistant questions about situations observed at stations and receive verbal responses. The assistant can provide answers both verbally and in text form, if necessary. The intelligent assistant being developed is adapted for use in the subway system, thanks to a well-trained model and has the ability to answer a wide range of questions about the subway and its stations. Usage of the digital twin in other areas is also possible after retraining the models.
3. **The proposed mobile digital assistant ensures uninterrupted operation over the internet**, unlike standard assistants, which, without internet access, become completely useless for certain issues or even stop working altogether.

Thus, the proposed solution will enable a transition from decision support systems using individual, specialized digital twins to systems based on the construction of multiple interconnected digital twins, and from individual intelligent assistants to new intelligent assistants that use interconnected digital twins to display detected events on 2D/3D object models and feature speech recognition and generation capabilities.

The proposed solution will improve the response speed of personnel in subway systems and other infrastructure facilities when events, including hazardous ones, occur. This will be achieved by rapidly processing data received from the subway facilities on a fog server using neural networks and visualizing the detected events on employees' smartphones.

The following performance indicators are proposed for evaluating the developed solution: the speed of event data acquisition by subway employees. The evaluation conditions include a stable TCP/IP access channel, an identical number of received events on time intervals, and identical time intervals.

To develop the digital twin, the authors formulated and solved the following tasks:

1. develop a digital twin of the subway and digital twins of the stations taking into account their interconnections, and develop a platform for their construction;
2. develop an intelligent assistant that uses the digital twin;
3. develop an application for a mobile platform.

5. Solution method

5.1. General description of the proposed solution

The proposed digital twin of the subway system includes multiple interconnected twins of different stations (Fig. 2). In its simplest form, the structure of the subway system's digital twin is two-level: on the first level the digital twins of individual stations are placed, and on the second, higher level, the subway system's digital twin, which includes the digital twins of individual stations, is located.

Events occurring at one station (e.g., a fire) are reflected in the digital twin of the subway system. The digital twins of the preceding station and the following station receive information about events (e.g., a fire), allowing subway staff and support services that works at other stations to quickly receive information about events occurring at neighboring stations.

Currently, many passenger collisions on station tracks are prevented only due to the attentiveness of the train driver. Failure of switch equipment and track deformation caused by wear can lead to disaster. Crowding on platforms can lead to crushes when entering train compartments, escalators, and exit stairs. In many of these cases, tragedies can be avoided by using the proposed interconnected digital twins, that take into account the states of neighboring stations, collect real-time data on subway facilities from various sources, and include an image recognition system that ensures high-accuracy event recognition.

When constructing the subway's digital twin and its station's digital twins, audio and sensor data are collected and processed in addition to video data, enabling the creation of datasets that enable highly accurate event detection. In new Tesla vehicles, sensors have been excluded from the autonomous driving system, and the road navigation system relies solely on information from surveillance cameras. Tesla's tests have shown that in some cases, the image recognition system is inaccurate, particularly in cases of deliberate sabotage (painting a road on a wall, etc.).

In the process of building the subway's digital twin for processing data received from different sources neural networks are used. Using neural networks helps prevent erroneous and inaccurate event detection at subway locations as by now neural networks have shown high efficiency in data processing, including image recognition.

The subway's digital twin contains the following objects: people, train cars, and robots (AI-powered car washers, which are planned for future implementation and use in the subway). The digital twin also contains stationary objects: kiosks, stairs, railway tracks, and utility rooms. The digital twin displays the movement of people, train arrivals, car door openings, vestibule closings, etc. based on pattern recognition using images from video cameras, as well as interactions between objects (the door's stuck, a breakdown, etc.) based on processing sensor data.

The digital twin also displays events such as fires, floods, escalator or car door failures, etc. The digital twin allows employees to conduct a chronological analysis of events with the additional ability to view video clips, sensor data, and listen to audio recordings, leveraging automated video analysis techniques [16].

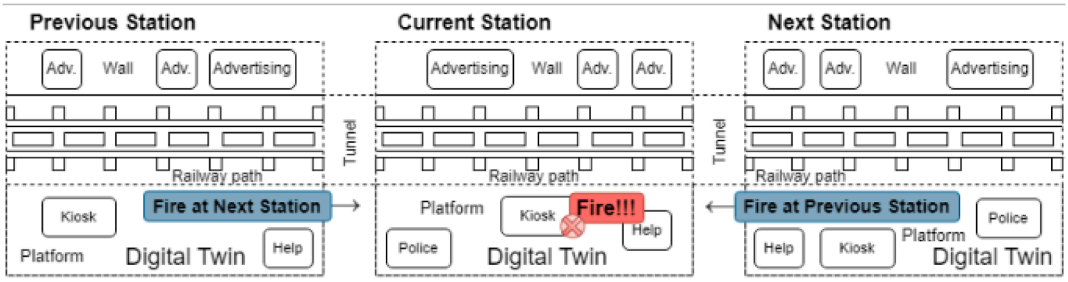


Fig. 2. Schematic representation of interconnected digital twins of subway stations.

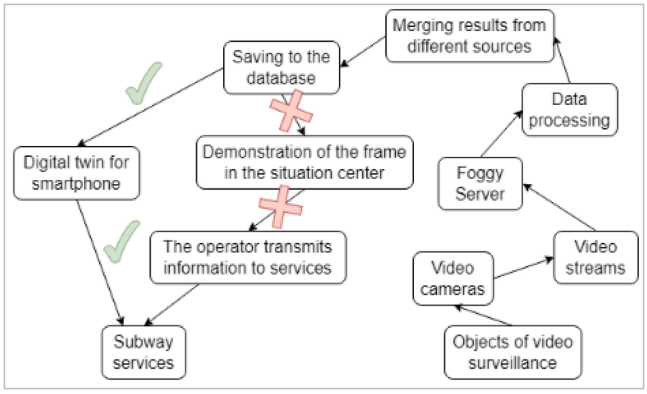


Fig. 3. General scheme of operation of the proposed digital twin.

The general operation of the digital twin is shown in Fig. 3. The objects of observation and the video cameras from which the video streams are received are placed at the bottom. On the **fog server**, the data is processed by the neural network and the results of data processing from various sources are subsequently fused.

Currently, instead of the sequence of steps discussed above, the following is performed: receiving data from video cameras via TCP/IP, transferring data from the video server to a single city data storage center, identifying objects and copying data to a secure server, displaying images in the situation center, and transmitting information to services by the operator.

The proposed solution is a comprehensive solution for identifying dangerous and emergency situations in subways with the ability to subsequently provide a detailed chronology of the occurrence and the course of events for interaction with law enforcement agencies and auxiliary services in the subway.

We benchmarked round-trip latency (Δt) across three topologies:

- Cloud-only: $\Delta t_{cloud} \approx 185$ ms (limited by bandwidth).
- Fog-environment (Implemented): $\Delta t_{fog} \approx 22$ ms.

This demonstrates that offloading computation to the fog environment ensures the near real-time synchronization required for safety-critical infrastructure.

5.2. Architecture of the decision support system

The architecture is presented in blocks with detailed information. Decision-making based on events requires an architecture (Fig. 4) consisting of two modules.

The first module is a “small digital twin” of a single station that is intended for the operator. The operator monitors safety at the subway station. Information for the operator is obtained from the digital twin. The digital twin analyzes incoming and previous information, and an event is generated for the operator. In Module 1 of the digital twin the situation at various stations is assessed. When new data is obtained a digital twin is rebuilt and an event is generated on the digital twin. All these steps are interconnected. Then, the event history is analyzed, and the best response to the event is determined. All these steps are inseparably linked.

Module 2 houses a “large digital twin” of the entire subway. The server processes data generating summary information on the events. The quality of the decision-making process depends on interconnected modules 1-2, exchanging information on the events and decisions made in response to them. Computational costs are minimal, since all the most complex operations are processed on the fog server.

The proposed architecture is distinguished by deep detailed processing of occurring events in real-time in combination with general processing of events that occur across all subway stations. Decision-making is based on the information received from monitored and at other neighboring stations. The architecture of the system is modular and block-based. Modules are decomposed into individual blocks and interconnections are defined for the blocks. Overall, the architecture is designed to generate well founded decisions and ultimately respond to events.

5.3. Platform for constructing a digital twin for the subway

Existing digital twin platforms generally don’t require programming; it is sufficient to have design skills. To construct a digital twin using these platforms it is required to place objects, build chains, and define connections with digital twin. This approach allows for the rapid development of digital twins.

However, existing platforms are designed for modeling specific objects in a specific domain. Not all platforms support image recognition from video cameras and the Internet of Things (smartphones, etc.), and many are very slow. Existing digital twin platforms also have limitations on the types of modeled objects and other design features of the models. They operate quite slowly when the number of objects is large.

Customized digital twins developed as binary applications (*.apk, *.exe, *.elf, etc.) operate several times faster, but require programming skills. Building a customized digital twin can take weeks, sometimes months, but it doesn’t have the limitations of digital twins created using off-the-shelf platforms.

To build digital twins the authors developed their own platform. The platform provides sets of standard elements (configurations for the subject area). The digital twins are developed as part of the digital twin platform. The digital twin platform allows for the definition of object movements and the customization of data acquisition from various sources (video cameras, sensors) to determine the movement of objects and their type.

When developing the platform, the core was programmed and ready-made standard components were used, eliminating the need for the developer to spend time drawing each component. A professionally design of a component and its integration into the platform can take 1-2 weeks. If the digital twin platform developer lacks graphic design skills and relevant education, the components will look unpresentable and out of place. The appearance of the elements is quite important.

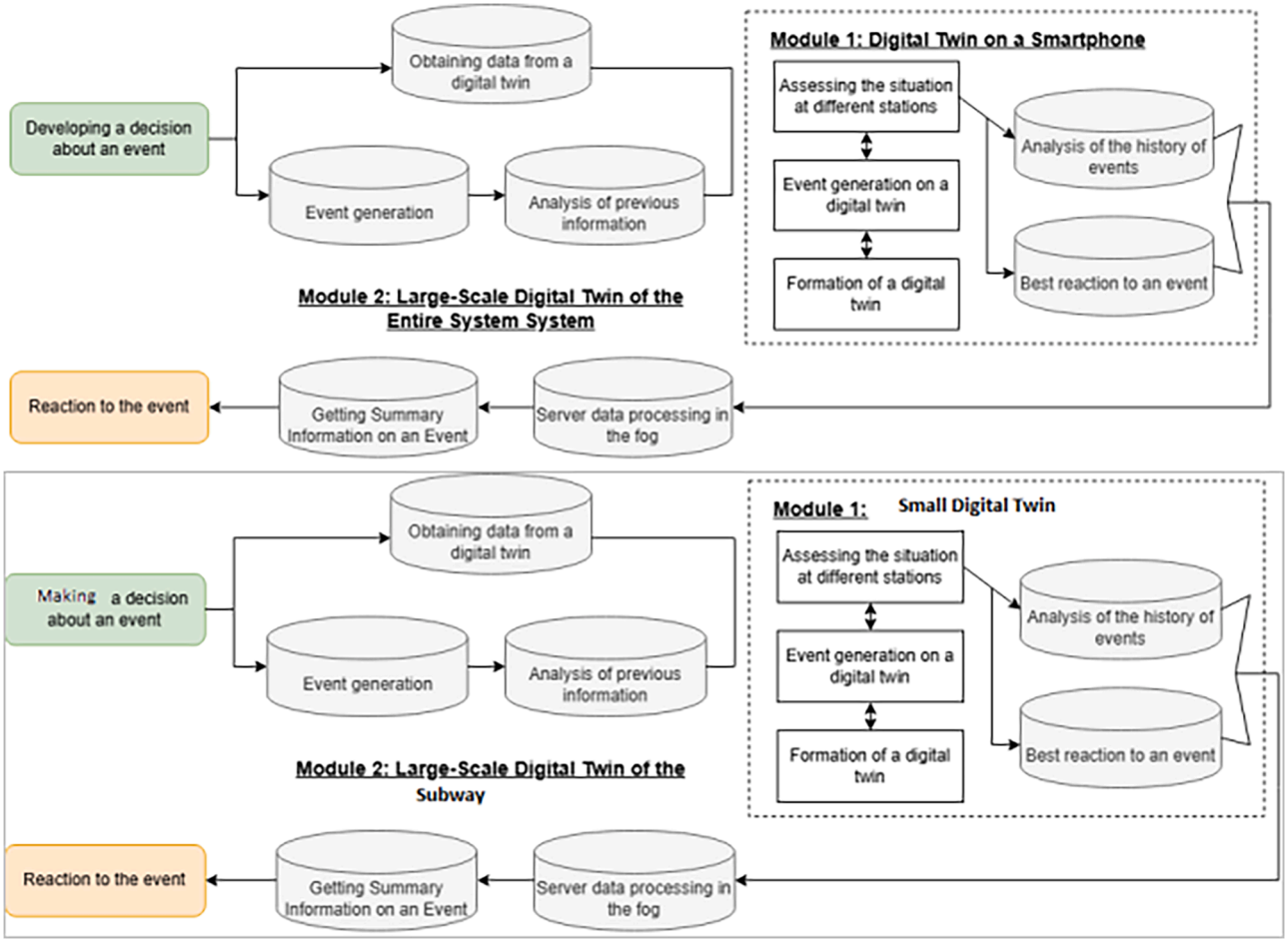


Fig. 4. Proposed architecture of the decision support system.

The developed platform also supports the work with sensors, microphones, and video cameras. This and other capabilities supported by the platform significantly speed up the creation of the digital twin. After creating all required objects and implementing them in the digital twin platform, creating a digital twin based on pre-defined elements becomes a very simple task. Adding a single new object to the model, containing elements from a set of pre-defined elements, takes 5–10 minutes.

The platform being developed includes a set of elements designed for subway facilities: kiosks, railway tracks, ticket offices, car wash machines, police posts, platforms, stairs, vestibules, escalators, station walkways, and other objects, including small ones, specifically needed in the subways. Each object has a detailed description. The platform and the digital twins created based on it form a single application.

When a mobile phone instance is launched, a connection to the fog server is established and GPS coordinates are transmitted. The system's server then generates a model based on the digital twin, taking into account the data about the user from whom the request was received, and sends the generated model to the user's app. The user sees a graphical model of the digital twin of the facility where he/she is currently located: a station, a lobby, a utility room, a depot, etc. When an event occurs, the graphical model automatically zooms in or out, displaying the location of the event, and a video is played. This is only possible with certain access rights. Event data is sent to users at the station where the event occurred and at neighboring stations. The operator also has the ability to navigate through the subway facilities in 2D/3D space and connect to video cameras to view the current situation at the facilities.

When using existing solutions, manual rebuilding of the digital twin is required when station architecture changes, new elements are added, or new stations are opened (commissioned). The proposed solution imposes no restrictions on the composition and parameters of subway objects, nor does it limit modeling to specific subway architecture and station objects. New elements can be added from a set of standard elements supported by the platform (kiosks, platforms, people, etc.). The new solution supports the use of not only specific camera types (AXIS P13/P33 series with DC-Iris and P-Iris) and sensors for data collection, but also other camera and sensor models.

5.4. Standard template for a digital twin of a station

New digital twins are built using a specially developed constructor and template, which represents a typical station digital twin. The digital twin constructor is a set of elements and relationships from the subject area. The elements are grouped based on their relevance: basic, advanced, additional, other, and selected. Elements are further divided into complex and simple.

Relationships between elements are configured depending of the connected elements. For example, a railway track can be connected to a tunnel, while people and traffic lights cannot, as they are elements of different entity types. The workspace for constructing digital twins is divided into the following main areas: the elements panel, the settings panel, the properties panel for the active (selected) element, and the main application menu for selecting panels, loading, and saving digital twins. Relationships between elements (and behavior for simulating events) are defined in a separate panel with detailed settings.

The digital twin template for a station includes a standard set of elements and links and, as a rule, requires minor modifications when

building new digital twins. Specifically, element and link properties can be changed: size, position, color, background, shadow, style, grid, text size, fill, line type and size, transparency, moveability, grouping with other elements, auto-size, text visibility, tooltips, text fields, font type and style, alignment, word wrap, and element border settings.

The final step in creating a digital twin based on a template is defining the inputs for the elements. Elements inputs can be connected with data sources or data can be generated using automatic data generation for the model inputs. To do this, it is necessary to open the menu for the selected (active) element, select the signal sources, data types, sampling frequency, intensity, modularity, and time interval. After all settings for the digital twin model of the station are completed, testing is performed. For testing a panel with digital twin launch and debug settings is used. After clicking the "Simulate" button, the digital twin is built, and the elements respond and interact.

5.5. Implementation of the digital twin of the subway

The following components were implemented for decision support systems based on digital twins of the subway:

- An intelligent assistant with the ability to visualize events on 2D/3D models and speech recognition capabilities;
- A server component for processing data about subway assets obtained from video cameras, microphones, and various sensors.

The mobile application, codenamed "Dt2DtSub-0.4.98a.apk," was developed using the GIGA IDE Community Edition 2024.1.1 development environment. The development environment supports Java (Level 4) and Kotlin (Level 5), and can compile and run programs on a high-performance cloud server hosted by GIGA. Kotlin Novator is suitable for development on Android and Linux. Using Kotlin, a level 5 language, greatly simplifies development for the mobile platform. The GigaIDE development environment is in no way inferior to IDEA/PyCharm from IntelliJ, but surpasses it in the number of additional plugins.

The ready-made APK application for Android-v.15 is located in the smartphone's file system and runs on the smartphone itself with the "developer mode" option enabled in the settings. The "Dt2DtSub" application connects to the fog server on which the twin is built depending on geolocation (GPRS) and is in the mode of receiving events from the server.

The Cisco UCS C480 ML M5 is used as a server in the developed system. The server processes video sequences. SubScript, which is adapted for intelligent video surveillance systems, in particular, subway systems, was used to develop the server component of the system. SubScript enables the storage of settings and statistics in a database, as well as the visualization of events on graphical models of objects. Event visualization is performed in applications installed on the employees' mobile phones. The HUAWEI Mate X6 is compatible with the mobile app.

The server and smartphone models used are adapted for machine learning and neural networks, which increases the speed of data processing when identifying objects and events. Data objects in the app are presented to users as 2D/3D models with zoom capabilities and the ability to view fine details, displaying data on the results of object and event recognition.

The intelligent assistant features voice assistant functions and speech synthesis capabilities, building upon robust multimodal interaction frameworks [17]. VOSK is used for speech recognition, SAPI5 is used for speech generation, and NAudio is used for working with audio streams. The Vosk solution is a ready-made solution written in Python. Unlike SpeechRecognition, the Vosk library does not require an internet connection and can operate without third-party services. Vosk supports over 25 languages, is lightweight, easily integrates with programming languages C#, JavaScript, Java, etc.), and runs on all popular operating systems (Windows, Linux, and macOS).

A preliminary acoustic evaluation of the VOSK-based intelligent assistant was conducted using pre-recorded operator commands overlaid with ambient subway noise (average 85 dB). Under these conditions, the system exhibited a Word Error Rate (WER) of 14.2%. While functional for basic commands, this error rate highlights a limitation in noisy subterranean environments, suggesting that future iterations will require aggressive acoustic noise suppression pre-processing.

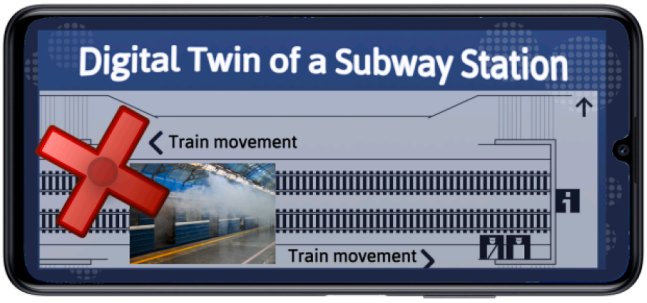


Fig. 5. Mobile application based on a digital twin providing data on neighboring stations.

Before the creation of the proposed intelligent assistant, there was no assistant capable of visualizing events on 2D/3D object models and integrated with a voice assistant. The assistant automatically refines the graphical model of the digital twin when identifying objects and events, and determines the display scale based on the level at which the event occurred. Furthermore, using a generative neural network, the assistant verbally describes the circumstances of the subway incident and the surrounding environment. The assistant suggests possible solutions, which can be accepted or rejected verbally. The developed app runs on smartphones used by operators at subway stations. Existing apps only run on Android and Linux, and cannot be compiled for iOS, macOS, or Windows Mobile.

For event detection, we utilized a modified YOLOv8 architecture optimized for edge devices. The model was fine-tuned on a custom dataset of $N = 12,500$ annotated frames. Training involved $E = 150$ epochs using the Adam optimizer with an initial learning rate $\alpha = 10^{-4}$ and batch size $B = 32$. The model achieved a mean Average Precision ($mAP@0.5$) of 92.4%.

6. Case study

The authors of the study developed a digital twin of the subway system, which includes digital twins of multiple interconnected stations. The mobile app (Fig. 5) is in line with global trends in mobile app development (developed for the popular Android 14 platform). The app is an intelligent assistant that displays information about the station where the operator is located and information about neighboring stations.

In the center of the app's main screen the real-world object (track, platform, etc.) and the detected event (fire, smoke) with its location are displayed. The application is designed to be as simple as possible, as working with simple elements improves the response time of subway personnel. The digital twin is displayed in a flat 2D space, but can be represented as a 3D model in case the "Dt2DtSub-0.4.98a.apk" app is opened on a device with a screen resolution of at least 800x600 pixels. The application detects the screen resolution upon launch and the 2D model is automatically converted into a 3D model, maintaining the proportions of the 3D space of one station.

The use of a digital twin and the mobile app with intelligent assistant functions for incident response at subway stations improved personnel response times by 17.4% (Table 3).

To ensure reliability, an Analysis of Variance (ANOVA) was conducted across $n = 50$ simulated trials per platform. The reduction in response time for Dt2DtSub was statistically significant ($F = 42.15$, $p < 0.001$). For a "Fire" event, the 95% confidence interval for response time was $[231.5, 242.5]$ seconds, compared to the REPEAT platform's $[272.8, 289.2]$ seconds.



Fig. 6. Comparative analysis of personnel response time reduction (%) using the proposed Dt2DtSub architecture versus existing digital twin platforms and baseline operations.

Table 3
Response time comparison with standard deviation ($\mu \pm \sigma$).

DT Name	Fire (sec)	Flooding (sec)	Collapse (sec)
REPEAT	281 ± 8.1	864 ± 12.4	2618 ± 45.2
XProtect	289 ± 9.3	874 ± 14.1	2654 ± 48.6
Dt2DtSub	$\mathbf{237 \pm 5.4}$	$\mathbf{714 \pm 9.8}$	$\mathbf{2157 \pm 33.1}$

Experimental studies have demonstrated the effectiveness of the developed digital twin, named Dt2DtSub. The creation and use of the subway digital twin in decision support systems has increased the response time of subway personnel and support services by an average of 18.6% across all events, compared to the current XProtect-based system, and by 17.4% compared to the most popular REPEAT platform.

The response time to the "Fire (Smoke)" event increased by 19.81%; to the "Flooding (Steaming)" event the response time increased by 24.63%; and to the "Collapse (Littering)" event the response time increased by 24.7% compared to the average response when using the existing digital twins.

Fig. 6 illustrates the comparative efficiency of the proposed solution. The graph is plotted against time (t). The x-axis shows the names of the digital twin platforms, and the y-axis represents the percentage of response time reduction relative to the proposed Dt2DtSub-0.4.98a solution.

To replicate the experiment, it is necessary to compile the mobile app and server component for building digital twins, collect statistics for three months of operation, open the logs in Myoffice Spreadsheet v.3.1 for Windows, enable Lua macros instead of VBA, apply filtering, round to decimal places, calculate the average value, autosum, and fill the resulting tables. The experiment was conducted on a Huawei Nova 13 Pro-12/512GB LTE Black smartphone with Android Linux 14.

7. Discussion

While the interconnected twin architecture demonstrates a 17.4% reduction in latency, this study relies heavily on simulated emergency scenarios. A primary limitation is the system's dependency on continuous TCP/IP connectivity; network degradation in deep subterranean environments could impede the fog-to-mobile data pipeline, requiring robust fallback protocols for physical, safety-critical deployment. Despite these constraints, the modular nature of the Dt2DtSub architecture offers high generalizability, allowing it to be scaled to other complex infrastructure networks such as airport terminals or underground mining facilities.

8. Conclusion

The authors proposed a digital twin of the subway system, built on the basis of interconnected digital twins of individual stations for decision support systems. We also developed a platform for building digital twins and an application for operators' mobile phones with intelligent assistant functions.

The developed mobile application functions as an intelligent assistant, displaying data about subway stations on smartphones while accounting for situations occurring at neighboring stations, and supports voice data processing. A key advantage is the high operating speed, ensured by the lightweight nature of the application and the offloading of heavy computation to the fog computing environment.

Experimental assessments demonstrated that the implementation of the Dt2DtSub architecture significantly accelerated emergency response protocols. Personnel response times improved by an average of 17.4% compared to the popular REPEAT platform. Specifically, the time required to contain fire and smoke incidents was reduced by 52 seconds compared to the current XProtect-based solution, and by roughly 47 seconds compared to the industry average. The absolute response times varied between a minimum of 267 seconds and a maximum of 307 seconds, validating the efficiency of offloading heavy computational tasks to the fog environment.

The proposed solution is applicable not only to subway systems but also to building models of other public transport assets: bus depots, trolleybuses, trams, and assets related to other network infrastructure (river transport and general rail service), where the performance of passenger company personnel needs to be improved.

Further developments include deeper integration with signaling and control systems to automatically stop mechanisms (trains, escalators, disconnecting live cables, etc.). Plans include enhancing the assistant's intelligence and support the intelligent analysis of complex information received from "Safe City" video cameras and the creation of event chronologies.

Future plans also include assessing the coordination of personnel actions, which assumes delegating responses to events from one service to another and can take considerable time, as well as the coordination of emergency response actions. The assistant is expected to include the following data in the future: information about stations closed for repairs and ongoing city events (for example, a heating main accident), as well

as the following features: asking questions and receiving answers in AI text mode, forwarding documents (regarding complaints about the subway, service quality, contracts, and commercial proposals), as well as enabling a task scheduler and calendar, and integrating with email and social media for receiving and sending information from the operator.

Future plans also involve measuring and evaluating response times to other events (panic, fights, conflicts, etc.) not discussed in this article. A website with support for multiple operating systems is also planned.

CRediT authorship contribution statement

Radhakrishnan Delhibabu: Writing – review & editing, Visualization, Validation, Software; **Nataly Zhukova:** Writing – original draft, Software, Resources, Project administration, Methodology, Funding acquisition; **Alexey Subbotin:** Investigation, Formal analysis, Data curation, Conceptualization.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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